Credit Portfolio Risk Monitoring

Historical credit-card default case study | UCI Default of Credit Card Clients (Taiwan) | October 2026

Research question

With capacity to review only 20% of accounts, can a statistical ranking concentrate more observed defaults in that review group than random selection? The analysis also explores how assumed default and recovery conditions affect a stylized portfolio-loss calculation.

Data and method

The UCI credit-card dataset contains 30,000 historical Taiwanese observations. A logistic regression model was trained on 75% of accounts and evaluated on a stratified random 25% holdout (7,500 accounts). Numerical features were standardized. No out-of-time validation was performed. Predicted probabilities were used to rank holdout accounts for a retrospective review-capacity exercise.

Model discrimination and prediction quality

Holdout measure	Value	Interpretation
ROC-AUC	0.716	Moderate discrimination between default and non-default accounts
Average precision (PR-AUC)	0.502	Above the 0.221 prevalence benchmark
Brier score	0.146	Mean squared error of predicted default probabilities
Observed default rate	22.12%	1,659 defaults among 7,500 holdout observations

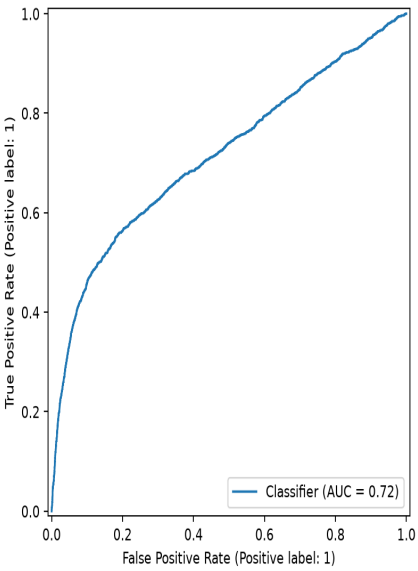


Figure 1. Holdout ROC curve. AUC rounds to 0.72 on the chart.

Review capacity and prioritization

Retrospective top-20% review rule

Accounts were sorted by predicted probability of default, highest first. The highest-ranked 1,500 accounts were selected for a hypothetical manual review, representing 20% of the 7,500-account holdout sample.

Metric	Observed result
Accounts selected for review	1,500 of 7,500 (20%)
Defaults in selected group	805
Total defaults in holdout	1,659
Share of defaults captured	48.52%
Default rate within selected group	53.67%
Lift versus random review	2.43x

Interpretation

Reviewing the highest-scored 20% of accounts would have covered 48.5% of the defaults observed in this historical holdout, versus an expected 20% under random selection. That is a 2.43-fold concentration. The finding supports the value of the risk ranking for retrospective prioritization, but does not establish that account review would prevent default or reduce realized losses.

Trade-offs and calibration

The selected review group still contains 695 accounts that did not default in the observed outcome period. Meanwhile, 854 observed defaults fall outside the selected group. Increasing review capacity could capture additional defaults at higher operational cost; decreasing capacity would require stricter prioritization. The calibration-by-decile plot shows differences between mean predicted probabilities and observed rates, including underprediction in the highest predicted-risk decile. Rankings therefore look more defensible than using individual predicted probabilities as validated financial forecasts.

SQL portfolio monitoring

Account-level holdout predictions were stored in SQLite and summarized by predicted-risk band. This allows comparison of observed default rates, predicted probabilities and credit-limit exposure proxies across portfolio segments. Risk bands support monitoring summaries, but the numerical cutoffs are illustrative rather than regulatory or institution-approved thresholds.

Decision implication: a risk team with fixed review capacity could use model ranking as one input to triage, subject to additional calibration, fairness, operational-cost and out-of-time testing.

Loss sensitivity and limitations

Illustrative loss framework

For each holdout account, an illustrative loss amount was computed as min(predicted PD x scenario multiplier, 100%) x credit limit x assumed LGD. The LIMIT_BAL / X1 credit-limit variable is denominated in New Taiwan dollars. It is **not** actual exposure at default (EAD), and the source does not contain recovery outcomes.

Scenario	PD multiplier	Assumed LGD	Illustrative loss (NT$)
Baseline	1.00x	60%	128.9 million
Downturn	1.25x	60%	160.9 million
Severe	1.60x	60%	204.6 million

The credit-limit total for the holdout is NT$1,257.2 million. The 60% LGD and PD multipliers are hypothetical sensitivity assumptions, not estimates validated on this population or tied to an economic forecast. These figures are illustrative modeled values rather than measured or accounting-recognized expected credit losses.

Methodological limitations

Population: Historical accounts from Taiwan cannot establish performance on Canadian borrowers or contemporary lending portfolios.

Validation: The test set is a stratified random holdout. It cannot demonstrate stability over later periods, through changing economic conditions or under population drift.

Calibration: Predicted and observed default rates diverge across predicted-risk deciles. Probability recalibration and independent validation would be necessary before loss estimation use.

Exposure and recovery: Credit limit is a proxy rather than EAD; LGD is assumed. Consequently the scenario totals should not be treated as actual or forecast portfolio losses.

Implementation: Fairness, privacy, model governance, operational cost, relevant legal requirements and human review procedures have not been evaluated.

Conclusion

The logistic regression baseline provides moderate default discrimination and useful retrospective concentration at limited review capacity: the highest-ranked 20% of accounts contained 48.5% of observed defaults. This supports a narrowly framed monitoring exercise, while the calibration and financial-loss assumptions rule out direct use as a validated lending or provisioning model.

Source and reproducibility

Yeh, I.-C., & Lien, C.-h. (2009), *The comparisons of data mining techniques for the predictive accuracy of probability of default of credit card clients*. Dataset distributed via UCI Machine Learning Repository, dataset ID 350. Analysis uses Python, pandas, scikit-learn, Matplotlib and SQLite; code and outputs are provided separately in the repository.